ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

DISTANCE LEARNING PROGRAMME
(Academic Session : 2024 - 2025)

JEE (Advanced)
AIOT
20-04-2025

JEE(Main + Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

Test Type : Full Syllabus

ANSWER KEY PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,C,D	A,C	C,D	A,C,D	A,C	A,B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	B	A	A		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	12.00	100.00	0.25	121.14 to 121.15	201.60	10.08
SECTION-II (ii)	Q.	7	8	9			
	A.	2	1	8			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C	A,B,C,D	A,C,D	A,B,D	A,B,C	A,B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	D	D	C	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.00	4.00	172.41	1.15	101.00	150.55
SECTION-II (ii)	Q.	7	8	9			
	A.	3	3	7			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,C	A,B,C,D	A,D	B,C	B,D	A,B,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	D	D	A		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.00	2.00	12.00	21.00	4.00	3.00
SECTION-II (ii)	Q.	7	8	9			
	A.	4	2	90			

HINT – SHEET

For More Material Join: @JEEAdvanced_2025

PART-1 : PHYSICS

SECTION-II (i)

6. Ans (10.08)

$$F = QE = Q\left(\frac{\rho R}{6\epsilon_0}\right)$$

$$= \frac{QR}{6\epsilon_0}\left(\frac{Q}{\frac{4\pi}{3}\left(R^3 - \frac{R^3}{8}\right)}\right) = \frac{Q^2}{7\pi\epsilon_0 R^2}$$

PART-1 : PHYSICS

SECTION-II (ii)

9. Ans (8)

For the first question, apply the Boltzmann equation, and

$$P = \sigma A T_s^4$$

where A is the surface area of the planet, and T_s the temperature at the center. Then

$$T_s = \left(\frac{P}{4\pi\sigma R^2}\right)^{\frac{1}{4}}$$

For the second question, it is reasonable to assume that the temperature depends on the distance form the center only. Then the definition of k gives for a spherical shell of thickness dr

$$k = \frac{\Delta Q}{\Delta t}\frac{1}{4\pi r^2}\frac{dr}{dT}.$$

The heat through the shell depends on the power radiated from within the shell. Since the planet is uniform, this depends on the volume according to

$$\frac{\Delta Q}{\Delta t} = P\frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi R^3} = P\frac{r^3}{R^3}$$

so that rearrangement yields

$$dT = \frac{P}{4\pi k R^3} r\, dr$$

Integrating between the center and the surface,

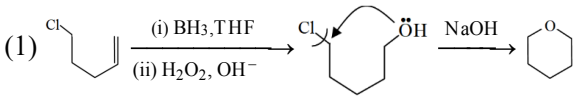
$$\Delta T = \frac{P}{8\pi k R},$$

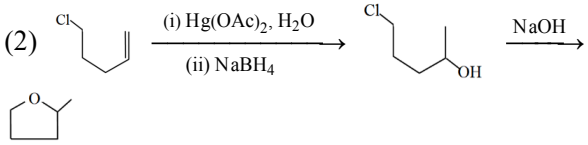
which could be used to find the temperature of the interior.

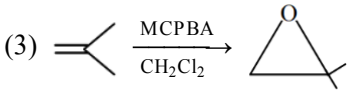
PART-2 : CHEMISTRY

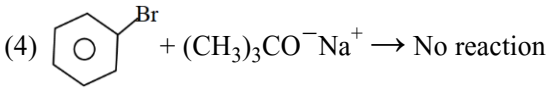
SECTION-I (i)

1. Ans (A,C)

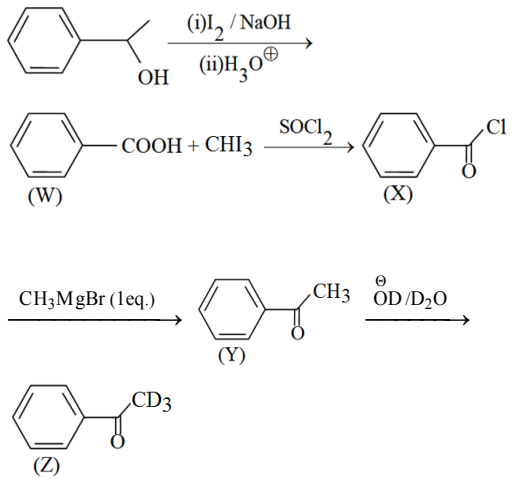
(1) 

(2) 

(3) 

(4) 

2. Ans (A,B,C,D)



3. Ans (A,C,D)

$$\underbrace{K_2Zn_3[\overset{+2}{Fe}(CN)_6]_2}_{\text{diamagnetic}} \xrightarrow{NaOH} K^+ + Zn(OH)_4^{2-} + [Fe(CN)_6]^{4-}$$

Synergic bonding b/w Fe^{+2} and $\bar{C}N$.

4. Ans (A,B,D)

X = $XeOF_4$, Y = XeO_2F_2, Z = XeO_3

5. Ans (A,B,C)

M = Cu

$$Cu^{2+} \xrightarrow{K_4[Fe(CN)_6]} \underset{\text{(Brown)}}{Cu_2[Fe(CN)_6]} \downarrow$$

$$E^\circ_{Cu^{+2}/Cu} = \text{'+'Ve}$$

For More Material Join: @JEEAdvanced_2025

6. **Ans (A,B,C)**

(A) pH = 5 + log 1 = 5 (Buffer solution)

(B) salt hydrolysis

(C) $[H^+] = \sqrt{CKa} = 10^{-5}$; pH = 5

(D) $[OH^-] = 2 \times 5 \times 10^{-6} = 10^{-5}$, pOH = 5, pH = 9

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (D)**

Rate of diffusion of methane

$= \frac{1}{16 \times 20}$ mole sec^{-1}

Rate of diffusion of hydrocarbon

$= \frac{\sqrt{20}}{M \times 40}$ moles sec^{-1}

Now, $\frac{r_{CH_4}}{r_{hyd}} = \sqrt{\frac{M_{hyd}}{16}}$

$\frac{\frac{1}{16\times20}}{\frac{\sqrt{20}}{M\times40}} = \frac{\sqrt{M}}{4} \Rightarrow \frac{40 \times M}{16 \times 20 \times \sqrt{20}} = \frac{\sqrt{M}}{4}$

$\Rightarrow \sqrt{M} = 2\sqrt{20}$

$\Rightarrow M = 80$ g mol^{-1}

0.125 m mol of (A) consumes 0.375 m mol H_2 . So,

(A) has three π-bonds

Therefore : molecular formula of A is C_6H_8

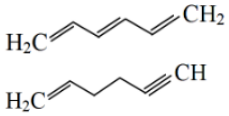
An open chain isomer of A can be

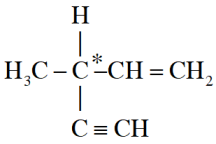
$CH_2 = CH - CH = CH - CH = CH_2$

A six membered stable monocyclic isomer of (A) can be  or 

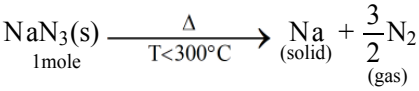
8. **Ans (D)**

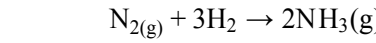
$\Rightarrow$ = M = 80 g mol^{-1}

 and 

 (Terminal alkynes responsed to tollen's test)

10. **Ans (D)**





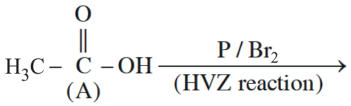


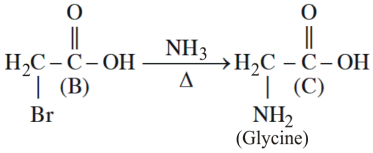
PART-2 : CHEMISTRY

SECTION-II (i)

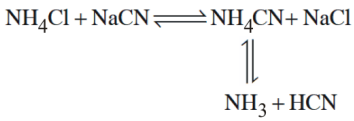
1. **Ans (2.00)**

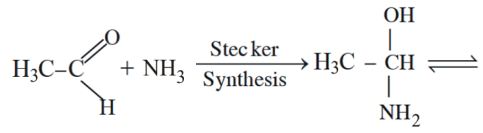
Acid (A) is ethanoic acid as it is an important constitutent of vineagar.

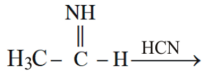


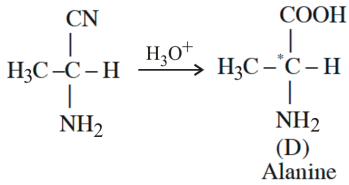


Also,









Now, since (C) has no chiral carbon. So, x = 0

And (D) has one asymmetric carbon, so y = 2

For More Material Join: @JEEAdvanced_2025

2. **Ans (4.00)**

Total number (z) of dipeptides which can be made by (C) and (D) are

gly - gly, ala - ala, gly - ala, ala - gly

Hence, z = 4

3. **Ans (172.41)**

$$Cv_{mix} = \left(\frac{n_1Cv_1 + n_2Cv_2}{n_1 + n_2}\right)$$

$$= \frac{\left(0.5 \times \frac{5R}{2}\right) + \left(0.5 \times \frac{5R}{2}\right)}{1}$$

$$Cv_{mix} = \frac{5R}{2}$$

$$Cp_{mix} = R + Cv_{mix}$$

$$= R + \frac{5R}{2}$$

$$= \frac{7R}{2}$$

For reversible adiabatic process

$$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$$

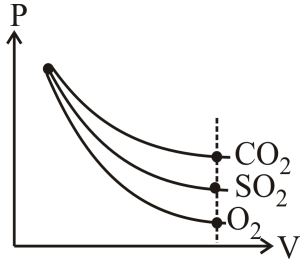
$$300(1)^{r-1} = t_2(4)^{\frac{7}{5}}$$

$$300 \times 1 = T_2(4)^{\frac{2}{5}}$$

$$300 = T_2 \times 1.74$$

$$T_2 = (172.41\ K)$$

4. **Ans (1.15)**



$X = O_2 \quad \gamma = \frac{9}{7} = 1.28$

$Y = CO_2 \quad \gamma = \frac{15}{13} = 1.15$

$Z = SO_2 \quad \gamma = \frac{7}{6} = 1.16$

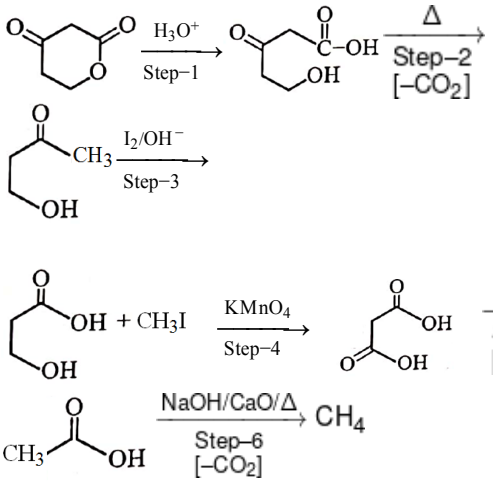
5. **Ans (101.00)**

$G = NaHCO_3$, $D = NH_3$

PART-2 : CHEMISTRY

SECTION-II (ii)

7. **Ans (3)**



8. **Ans (3)**

$\Delta T_b = i.K_b.m$

$0.08 = i \times 0.5 \times \frac{2}{20}$

$\frac{0.08}{0.5} \times \frac{20}{2} = i$

$i = 1.6$

$\Lambda_m = \Lambda_m^{\circ}.C.\alpha.$

$= 500\frac{Scm^2}{mole} \times \frac{2}{20}\frac{mole}{L} \times 0.6$

$\Lambda_m = 30\frac{Scm^2}{1000cm^3} = \frac{3}{100}Scm^{-1}$

$\Lambda_m = 3\ Sm^{-1}$

9. **Ans (7)**

$BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 \downarrow +2NaCl$

0.2 M 0.1 M ppt

0.04 mole 0.02 mole 0.08 mole

$\pi_{Na_2SO_4} = i_1C_1RT \qquad \pi_{NaCl} = i_2C_2RT$

$\pi_{Na_2SO_4} + \pi_{NaCl} = (i_1C_1 + i_2C_2).RT$

$= 3 \times \left(\frac{0.02}{0.8}\right) + 2\left(\frac{0.08}{0.8}\right).R \times T$

$= (0.075 + 0.2)RT$

$= 0.275 \times 0.0821\ atm \times 300$

$= 6.99$

$= 7\ atm$

For More Material Join: @JEEAdvanced_2025

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B,C)

$xf(y) < yf(x)\ \forall\ 0 < x < y < 1$

$\Rightarrow \quad \frac{f(y)}{y} < \frac{f(x)}{x}\ \forall x < y$

$\Rightarrow \quad g(x) = \frac{f(x)}{x}$ is decreasing

$\Rightarrow \quad g'(x) < 0 \Rightarrow xf'(x) < f(x)$

$\Rightarrow \quad f'(x) < \frac{f(x)}{x} \forall\ x, 0 < x < 1$

Hence, $f'(x) < g(x) \forall\ x, 0 < x < 1$

Hence, f '(x) is less than minimum value of g(x) which is g (1)

$\therefore\ f'(x) < f(1)$

$x\ f'(x) < f(x) \Rightarrow \int_0^1 x\, f'(x)\, dx < \int_0^1 f(x)\, dx$

$\Rightarrow x\ f(x)\ |_0^1 - \int_0^1 f(x)\ dx < \int_0^1 f(x)\ dx$

$\Rightarrow \quad f(1) < 2\int_0^1 f(x)\ dx$

2. Ans (A,B,C,D)

$|z_1\ z_2| = \left|\frac{c}{a}\right| = 1, \quad |z_1 + z_2| = \left|\frac{-b}{a}\right| = 1$

$\Rightarrow\ (z_1 + z_2) \times (\bar{z}_1 + \bar{z}_2) = 1$

$\Rightarrow (z_1 + z_2)\left(\frac{1}{z_1} + \frac{1}{z_2}\right) = 1$

$\Rightarrow \quad (z_1 + z_2)^2 = z_1\ z_2$

$\Rightarrow \quad \left(\frac{-b}{a}\right)^2 = \frac{c}{a} \Rightarrow b^2 = ac$

$|z_1 + z_2| = |z_1||1 + e^{i\theta}| = 2\cos\frac{\theta}{2} = 1 \Rightarrow \theta = \frac{2\pi}{3}$

$PQ = |z_2 - z_1| = |z_1||e^{i\theta} - 1| = 2\sin\frac{\theta}{2}$

$= 2\sin\frac{\pi}{3} = \sqrt{3}$

3. Ans (A,D)

(a) $T_n^3 + T_n = T_{n+1} - T_n$

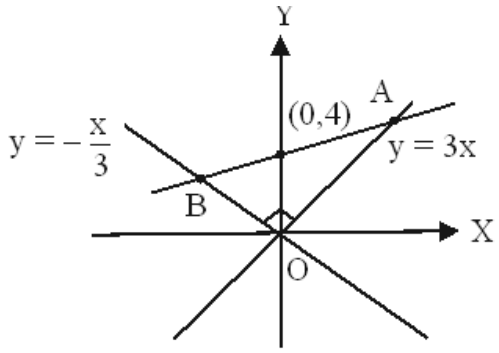
$\sum_{n=1}^{100}(T_n^3 + T_n + 1) = T_{101} - T_1 + 100 = T_{101} + 99$

(c) $T_n^2 + 2 = \frac{T_{n+1}}{T_n}$

$\prod_{n=1}^{100}(T_n^2 + 2) = \prod_{n=1}^{100}\frac{T_{n+1}}{T_n} = \frac{T_{101}}{T_1} = T_{101}$

(d) $\prod_{n=1}^{100}\left(\frac{T_{n+1}}{T_n} - T_n^2\right) = 2^{100}$

4. Ans (B,C)



$(3x - y)(x + 3y) = 0$

$y = 3x, y = \frac{-x}{3}$

(a) $m_1 + m_2 = 3 - \frac{1}{3} = \frac{8}{3}$

(b) $am_1 + m_2 = 3\left(-\frac{1}{3}\right) + 3 = 2$

(c) $y = 2x + 4$

$A \equiv (4, 12), B = \left(\frac{-12}{7}, \frac{4}{7}\right)$

$\text{Area}(\Delta AOB) = \frac{1}{2} \times \sqrt{16 + 144} \times \sqrt{\frac{144 + 16}{49}} = \frac{160}{2 \times 7} = \frac{80}{7}$

5. Ans (B,D)

$(A^2 - 2I)B = 0 \Rightarrow A^2 = 2I$ and

$B = \text{adj}\ A = |A|.\ \frac{A}{2} = -\sqrt{2}A \{\because |A| = -2\sqrt{2}\}$

$AB = A\ (\text{adj}\ A) = |A|.\ I$

$\text{tr.}\ (AB) = 3|A| = -6\sqrt{2}$

and $\det.\ (A - \sqrt{2}B) = \det.\ (A + 2A) = \det.\ (3A)$

$= 27(-2\sqrt{2}) = -54\sqrt{2}$

For More Material Join: @JEEAdvanced_2025

6. Ans (A,B,D)

$x^3 - 3x + \lambda = 0$ let root = α, β, γ

$\therefore \alpha+\beta+\gamma = 0,\ \alpha\beta+\beta\gamma+\gamma\alpha = -3,\ \alpha\beta\gamma = -\lambda$

$\therefore \alpha^2+\beta^2+\gamma^2 = 6,\ (\alpha,\beta,\gamma \in I)$

$\therefore$ Possibilities are

$$\left.\begin{matrix}\alpha=2,\beta=-1,\gamma=-1\\ \alpha=-2,\beta=1,\gamma=1\end{matrix}\right\} \Rightarrow \lambda = 2 \text{ or } -2 \Rightarrow (\lambda_1)^2+(\lambda_2)^2 = 8$$

$\therefore h(x) = x^3 - 3x + 4$

Required area

$$= \int_2^5 h(x)\,dx = \int_2^5 (x^3-3x+4)\,dx = \frac{531}{4}$$

And $\int_1^3 (x^3-3x)\,dx + \int_2^{22} g(x)\,dx = \int_1^3 (x^3-3x+4)\,dx + \int_2^{22} g(x)\,dx - 8$

$= 66 - 2 - 8 = 56$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (B)

a, b and c are in A.P.

$\therefore \sqrt{ac} < b$

$$f(x) = \lim_{n\to\infty} \frac{-(e^x)^n + x^2 + f}{2e^x(e^x)^n + x + d} = \begin{cases} \frac{x^2+f}{x+d} & , \quad x<0 \\ \frac{-1+f}{2+d} & , \quad x=0 \\ \frac{-1}{2e^x} & , \quad x>0 \end{cases}$$

For f(x) to be continuous

$$\frac{f}{d} = \frac{-1+f}{2+d} = \frac{-1}{2}$$

$\therefore 2f + d = 0 \Rightarrow 2f + d + 1 = 1$

8. Ans (D)

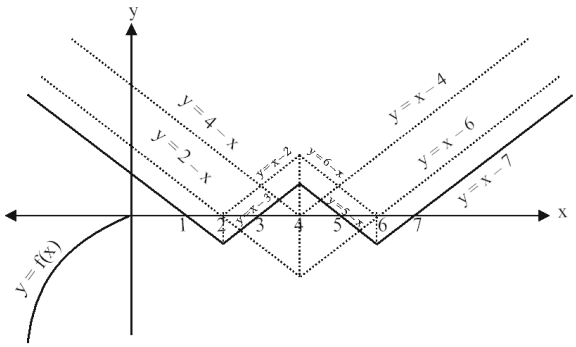
a, b and c are in G.P.

$$f(x) = \lim_{n\to\infty} \frac{x^2+f}{2e^x(e^x)^n + x + d} = \begin{cases} \frac{x^2+f}{x+d} & , \quad x<0 \\ \frac{f}{2+d} & , \quad x=0 \\ 0 & , \quad x>0 \end{cases}$$

For f(x) to be continuous

$$\frac{f}{d} = \frac{f}{2+d} = 0 \qquad \Rightarrow \ f = 0$$

$f(x) = ||x-4|-2|-1$



No of solution = 4

9. Ans (D)

$$P\left(\frac{A}{B_E}\right) = \frac{P(A \cap B_E)}{P(B_E)}, \text{ where A = sum is 4.}$$

And $P(B_E) = P(B_2) + P(B_4) + \ldots\ldots = \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \ldots\ldots\infty$

$$= \frac{\frac{1}{4}}{1-\frac{1}{4}} = \frac{1}{4}\cdot\frac{4}{3} = \frac{1}{3}$$

$P(A \cap B_E) = P(A \cap B_2) + P(A \cap B_4)$

$$= P(B_2)P\left(\frac{A}{B_2}\right) + P(B_4)P\left(\frac{A}{B_4}\right)$$

$$= \frac{1}{4}\left(\frac{3}{36}\right) + \frac{1}{16}\left(\frac{1}{6^4}\right) = \frac{(4.3.6^2+1)}{16.6^4} = \frac{433}{16.6^4}$$

Hence $P\left(\frac{A}{B_E}\right) = \frac{433}{16.6^4}.3 = \frac{433}{32.216} \cong \frac{1}{16}$

10. Ans (A)

$n(S) = 6 \times 6 \times 6 = 6^3 = 216$

Now greatest number is 4, so atleast one of the dice shows up 4.

$\therefore n(A) = 4^3 - 3^3 = 37$

Hence $P(A) = \frac{37}{216}$

For More Material Join: @JEEAdvanced_2025

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (1.00)

a, c, b are in H.P, $\Rightarrow \frac{2}{c} = \frac{1}{a} + \frac{1}{b}$

Family of lines

$\frac{x}{a} + \frac{y}{b} + \frac{1}{c} = 0 \Rightarrow \frac{x}{a} + \frac{y}{b} + \frac{1}{2}\left(\frac{1}{a} + \frac{1}{b}\right) = 0$

$\Rightarrow \frac{1}{a}\left(x + \frac{1}{2}\right) + \frac{1}{b}\left(y + \frac{1}{2}\right) = 0$

$\Rightarrow$ centre of ellipse $\equiv \left(\frac{-1}{2}, \frac{-1}{2}\right)$

AC : $x^2 + y^2 + 2px + 2qy - 8p + \frac{1}{2} = 0$

$\Rightarrow p = \frac{1}{2} : q = \frac{1}{2}$; radius of AC = 2

$\therefore$ ℓ (major axis) = 2a = 4

$\Rightarrow$ We know, harmonic mean of segment of focal chord is equal to semi latus rectum.

semi latus rectum = $\frac{2.(1 \times 2)}{(1+2)} = \frac{4}{3}$

2. Ans (2.00)

a,c, b are in H.P, $\Rightarrow \frac{2}{c} = \frac{1}{a} + \frac{1}{b}$

Family of lines :

$\frac{x}{a} + \frac{y}{b} + \frac{1}{c} = 0 \Rightarrow \frac{x}{a} + \frac{y}{b} + \frac{1}{2}\left(\frac{1}{a} + \frac{1}{b}\right) = 0$

$\Rightarrow \frac{1}{a}\left(x + \frac{1}{2}\right) + \frac{1}{b}\left(y + \frac{1}{2}\right) = 0$

$\Rightarrow$ centre of ellipse $\equiv \left(\frac{-1}{2}, \frac{-1}{2}\right)$

AC : $x^2 + y^2 + px + 2qy - 8p + \frac{1}{2} = 0$

$\Rightarrow p = \frac{1}{2} : q = \frac{1}{2}$; radius of AC = 2

$\therefore$ *l* (major axis) = 2a = 4

$\Rightarrow$ We know, harmonic mean of segment of focal chord is equal to semi latus rectum.

Semi latus rectum

$= \frac{2.(1 \times 2)}{(1+2)} = \frac{4}{3}$

$\therefore$ latus rectum, $L = \frac{8}{3} = 2.67$

[L] = [2.67] = 2

3. Ans (12.00)

Put $x = \cos\theta$

$1 - x^2 = \sin^2\theta. \Rightarrow \sqrt{x^2 - 1} = i\sin\theta$

$\left(x + \sqrt{x^2-1}\right)^{10} + \left(x - \sqrt{x^2-1}\right)^{10}$

$= (\cos\theta + i\sin\theta)^{10} + (\cos\theta - i\sin\theta)^{10}$

$= 2\cos 10\theta = 2f_{10}(x) = \lambda f_k(x)$

$\therefore$ $\lambda = 2; k = 10 \Rightarrow \lambda + k = 12.$

4. Ans (21.00)

$f_6(\cos x) = \cos 6x$

we know,

$(\cos x + i\sin x)^6 = \cos 6x + i\sin 6x$

Equating real part both sides :

$\cos 6x = {}^6C_0\cos^6 x - {}^6C_2\cos^4 x\sin^2 x + {}^6C_4\cos^2 x\sin^4 x - {}^6C_6\sin^6 x$

$f_6(\cos x) = \cos 6x = \cos^6 x - 15\cos^4 x(1 - \cos^2 x) + 15\cos^2 x(1 - \cos^2 x)^2 - (1 - \cos^2 x)^3$

$\therefore f_6(x) = x^6 - 15x^4(1 - x^2) + 15x^2(1 - x^2)^2 - (1 - x^2)^3$

$\therefore p = 32 ; q = -48; r = 18, s = -1.$

$\therefore 7(p + q + r - s) = 7(32 - 48 + 18 + 1) = 7 \times 3 = 21$

5. Ans (4.00)

Equation of family of planes containing the line of intersection of P_1 and P_2 is given by $P_1 + \lambda P_2 = 0$

$\Rightarrow (1 + \lambda)x + (1 - 2\lambda)y + (2 + \lambda)z - 3 - 4\lambda = 0$

.....(1)

Projection of length of line segment AB is greatest, then AB must be parallel to plane (1)

$\therefore$ drs of AB = 2, – 1,3.

$2(1 + \lambda) - 1(1 - 2\lambda) + 3(2 + \lambda) = 0 \Rightarrow \lambda = -1$

$\therefore P_3 : 3y + z + 1 = 0. \Rightarrow a = 0, b = 3, c = 1$

For More Material Join: @JEEAdvanced_2025

6. Ans (3.00)

Projection of length of line segment is least if AB is perpendicular to plane (1).

$$\frac{1+\lambda}{2} = \frac{1-2\lambda}{-1} = \frac{2+\lambda}{3} \Rightarrow \lambda = 1$$

$\therefore$ Equation of plane P_4 is

$2x - y + 3z = 7. \Rightarrow a = 2, b = -1, c = 3$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (4)

$$\frac{1}{5}\left[\frac{1}{3}+\frac{1}{6}+\frac{1}{10}+\frac{1}{15}+\ldots\ldots\infty\right]$$

$$\frac{2}{5}\left[\frac{1}{6}+\frac{1}{12}+\frac{1}{20}+\frac{1}{30}+\ldots\ldots\infty\right]$$

$$\frac{2}{5}\left[\frac{(3-2)}{6}+\frac{(4-3)}{12}+\frac{(5-4)}{20}+\ldots\ldots\infty\right]$$

$$\frac{2}{5}\times\frac{1}{2} = \frac{k}{20} \Rightarrow k = 4$$

8. Ans (2)

$$\frac{a}{\cos(200^\circ)} - \frac{c\sin(200^\circ)}{\cos(200^\circ)} = d \quad \ldots\ldots\ldots\ldots (1)$$

$$\frac{b}{\cos(200^\circ)} + \frac{d\sin(200^\circ)}{\cos(200^\circ)} = c \quad \ldots\ldots\ldots\ldots\ldots (2)$$

$a = d\cos(200°) + c\sin(200°)$ ..... (3)

$b = c\cos(200°) - d\sin(200°)$ ..... (4)

$\therefore a^2 + b^2 = c^2 + d^2$

$\therefore (a^2 + b^2 + c^2 + d^2) = 2(c^2 + d^2)$

$ac = dc\cos(200°) + c^2\sin(200°)$

$bd = cd\cos(200°) - d^2\sin(200°)$

Also _____

$ac - bd = (c^2 + d^2)\sin(200°)$

$$\text{So}\frac{a^2+b^2+c^2+d^2}{ac-bd} = \frac{2(c^2+d^2)}{(c^2+d^2)\sin(200^\circ)} = 2\,\text{cosec}\,(200^\circ)$$

$\therefore p = 2$

9. Ans (90)

$$I_n = \left(\frac{e^{-x}(\sin x)^n}{-1}\right)\Big|_0^\infty + n\int_0^\infty [(\sin x)^{n-1}\cos x e^{-x}dx$$

$$= 0 + n\left(\frac{(\sin x)^{n-1}\cos x\, e^{-x}}{-1}\right)\Big|_0^\infty$$

$$+ n\int_0^\infty [(\sin x)^{n-1}(-\sin x) + (\cos x)(n-1)\cos x(\sin x)^{n-2}\cos x]e^{-x}dx$$

$$= n\int_0^\infty (-(\sin x)^n + (n-1)(1-\sin^2 x)(\sin x)^{n-2})e^{-x}dx$$

$$= \frac{n(n-1)}{n^2+1}I_{n-2}$$

$$101\frac{I_{10}}{I_8} = 101\times\frac{10\times 9}{10^2+1} = 90.$$

For More Material Join: @JEEAdvanced_2025